Polar Curves and Derivatives

Section 1: Slopes of Tangent Lines to Polar Curves

1 Which formula gives the slope $\frac{dy}{dx}$ of the tangent line to the polar curve $r = f(\theta)$?

a. $\frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$

b. $\frac{dr}{d\theta}$

c. $\frac{r\sin\theta}{r\cos\theta}$

d. $\frac{\frac{dr}{d\theta}\cos\theta}{\frac{dr}{d\theta}\sin\theta}$

2 Find the slope of the tangent line to the circle $r = 2\cos\theta$ at $\theta = \frac{\pi}{4}$.

a. -1

b. 1

c. 0

d. Undefined

3 What is the slope of the tangent line to the cardioid $r = 1 + \sin\theta$ at $\theta = 0$?

a. 1

b. -1

c. 0

d. Undefined

Section 2: Horizontal and Vertical Tangents on Polar Curves

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4 Horizontal tangent lines to a polar curve occur where which condition is met?

a. $\frac{dy}{d\theta} = 0$ and $\frac{dx}{d\theta} \neq 0$

b. $\frac{dr}{d\theta} = 0$

c. $\frac{dx}{d\theta} = 0$

d. $r = 0$

5 Vertical tangent lines occur where which condition is met?

a. $\frac{dx}{d\theta} = 0$ and $\frac{dy}{d\theta} \neq 0$

b. $\frac{dr}{d\theta} = 0$

c. $\frac{dy}{d\theta} = 0$

d. $r = 1$

6 At the pole ($r = 0$), what is the slope of the tangent line to the curve $r = f(\theta)$ when $f(\alpha) = 0$?

a. $\tan\alpha$

b. $\cot\alpha$

c. 0

d. Undefined

Section 3: Rate of Change of r and Distance from Origin

7 For the polar curve $r = 3 - 2\cos\theta$, what is $\frac{dr}{d\theta}$ at $\theta = \frac{\pi}{2}$?

a. 2

b. -2

c. 0

d. 3

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8 If $\frac{dr}{d\theta} > 0$ for a polar curve, what does this indicate about the distance from the pole?

a. The distance from the origin is increasing as θ increases

b. The distance is decreasing

c. The curve is concave up

d. The curve is a line

9 At what angle $\theta \in [0, 2\pi)$ is the distance from the origin maximized for $r = 2 + 3\cos\theta$?

a. $\theta = 0$

b. $\theta = \pi$

c. $\theta = \frac{\pi}{2}$

d. $\theta = \frac{3\pi}{2}$

Section 4: Arc Length of Polar Curves

10 Which integral represents the arc length of the polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$?

a. $\int_{\alpha}^{\beta} \sqrt{r^2 + (\frac{dr}{d\theta})^2}\, d\theta$

b. $\int_{\alpha}^{\beta} \sqrt{1 + (\frac{dr}{d\theta})^2}\, d\theta$

c. $\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$

d. $\int_{\alpha}^{\beta} r d\theta$

11 Find the total perimeter of the circle $r = 4\cos\theta$ for $\theta \in [0, \pi]$.

a. 4π

b. 8π

c. 2π

d. 16π

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12 Set up the arc length of the cardioid $r = 1 + \cos\theta$ from $\theta = 0$ to $\theta = 2\pi$.

a. $\int_0^{2\pi} \sqrt{2 + 2\cos\theta}\, d\theta$

b. $\int_0^{2\pi} (1 + \cos\theta) d\theta$

c. $\int_0^{2\pi} \sqrt{1 + \cos^2\theta}\, d\theta$

d. $2\int_0^{\pi} (1 + \sin\theta) d\theta$

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